Integration

Basics of Integration

Integration ≡ Anti-Differentiation

Table of derivatives

- $\frac{d}{dx} x^n = n x^{n-1}$
- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \ln x = \frac{1}{x}$
- $\frac{d}{dx} \sin x = \cos x$
- $\frac{d}{dx} \cos x = -\sin x$
- $\frac{d}{dx} \tan x = \sec^2 x$
- $\frac{d}{dx} \cot x = -\csc^2 x$
- $\frac{d}{dx} \sec x = \sec x \tan x$
- $\frac{d}{dx} \csc x = -\csc x \cot x$

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- $\frac{d}{dx} \cot x = -\csc^2 x$
- $\frac{d}{dx} \sec x = \sec x \tan x$
- $\frac{d}{dx} \csc x = -\csc x \cot x$

Notation of Integrals

$\int f(x)\,dx$: The indefinite integral of $f(x)$ with respect to x i.e. a function whose derivative is $f(x)$

If $\frac{d}{dx} F(x) = f(x)$ then $\int f(x)\,dx = F(x)$

eg: $\frac{d}{dx} x^2 = 2x \quad \therefore \int 2x\,dx = x^2 + c$ $\qquad \frac{d}{dx} c = 0$

eg: $\frac{d}{dx} \sin x = \cos x \Rightarrow \int \cos x\,dx = \sin x + c$

eg: $\frac{d}{dx} e^x = e^x \Rightarrow \int e^x\,dx = e^x + c$

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eg: $\int \frac{3}{2} x^{1/2}\,dx = x^{3/2} + c$ $\qquad \frac{d}{dx} x^{3/2} = \frac{3}{2} x^{1/2}$

Rules to calculate integrals

$$\int c \cdot f(x)\,dx = c \cdot \int f(x)\,dx$$

$$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

Basic integrals

$$\int 1\,dx = x + c \qquad \int a\,dx = ax + c$$

$$\int x^n\,dx = \frac{x^{n+1}}{(n+1)} + c$$

$$\int e^x\,dx = e^x + c \qquad \int \frac{1}{x}\,dx = \ln x + c$$

$$\int a^x\,dx = \frac{a^x}{\log a} + c$$

$$\int \sin x = -\cos x + c$$
$$\int \cos x = \sin x + c$$
$$\int \sec^2 x = \tan x + c \qquad \int \sec x\,dx = \log|\sec x + \tan x| + c$$
$$\int \operatorname{cosec}^2 x = -\cot x + c \qquad \int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + c$$
$$\int \tan x = \log|\sec x| + c$$
$$\int \cot x = \log|\sin x| + c$$

Evaluating integrals

Chain Rule:

In differentiation, $y = f(u)$, $u = g(x)$

$y = f(g(x))$

then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \quad \text{or} \quad \frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

E.g. $\frac{d}{dx} \sin 3x = \frac{d \sin(3x)}{d(3x)} \cdot \frac{d\,3x}{dx} = (\cos 3x) \cdot (3)$

In integration, $\int f'(g(x)) \cdot g'(x)\,dx = f(g(x)) + c$

e.g. $\int 3 \cdot e^{3x}\,dx$

(let $3x = t$, $3\,dx = dt$)

$= \int e^t\,dt$

$= e^t + c = e^{3x} + c$

e.g. $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$

let $\sqrt{x} = t$

$x^{1/2} = t$

$\frac{1}{2} x^{-1/2}\,dx = dt$

$\frac{dx}{\sqrt{x}} = 2\,dt$

$= \int \sin t \cdot 2\,dt$

$= 2 \int \sin t\,dt$

$= -2\cos t + c$

$= -2\cos\sqrt{x} + c$

Integration by parts:

$$\int u \cdot v\,dx = u \int v\,dx - \int \left(u' \cdot \int v\,dx\right) dx \qquad u = f(x),\ g = f(x)$$

e.g. $\int \underset{u}{x} \cdot \underset{v}{e^x}\,dx$

$= x \int e^x - \int 1 e^x\,dx = xe^x - e^x + c$

ILATE: Inverse Trig. Func^{n}s — Logarithmic func^{n}s — Algebraic func^{n}s — Trigonometric func^{n}s — Exponentials

in increasing order of precedence for v or priority from left for u.

$u\int v\,dx - \int(u'\int v\,dx)\,dx$

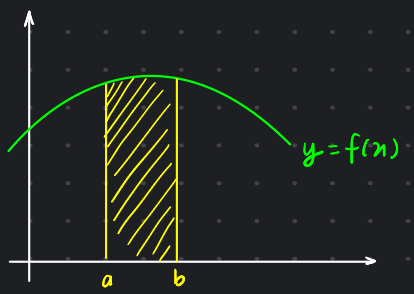
e.g. $\int \log x\,dx = \int \underset{u}{\log x}\cdot\underset{v}{1}\,dx$

$= \log x \cdot x - \int \frac{1}{x}\cdot x\,dx$

$= x\log x - x + C$

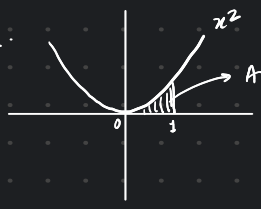
$= x(\log x - 1) + C$

Definite integrals



$\int_a^b f(x)\,dx = F(b) - F(a)$ where $\int f(x)\,dx = F(x) + C$

e.g.



Area $= \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

e.g. $\int_2^4 \frac{1}{x}\,dx = [\ln x]_2^4 = \ln 2$

$u\int v\,dx - \int(u'\int v\,dx)\,dx$

e.g. $\int_0^1 x\cdot e^{x^2}\,dx$

$x^2 = t$

$2x\,dx = dt$

$x\,dx = \frac{dt}{2}$

$\int x e^{x^2}\,dx$

$= \int e^t \frac{dt}{2}$

$= \frac{1}{2}e^t + C$

$= \frac{1}{2}e^{x^2} + C$

$\int_0^1 x e^{x^2}\,dx = \left[\frac{1}{2}e^{x^2}\right]_0^1$

$= \frac{1}{2}(e-1)$

Properties of definite integrals

$\int_a^a f(x)\,dx = 0$

If $a \le b \le c$, then

$\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$

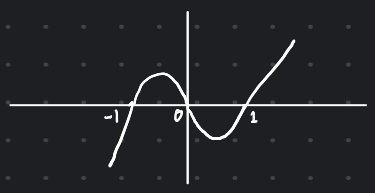
$\int_a^b c\cdot f(x)\,dx = c\int_a^b f(x)\,dx$

$\int_a^b (f(x)+g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

Definite integrals do not always represent the area under the curve.
The represent the area above the X-axis.

$f(x) = x(x^2-1)dx$



$$\int f(x)\,dx = \frac{x^4}{4} - \frac{x^2}{2} + C$$

$$\int_{-1}^{0} f(x)\,dx = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4}$$



$$\int_{0}^{1} f(x)\,dx = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$



$$\int_{-1}^{1} f(x)\,dx = 0$$



e.g. $\int_1^5 \frac{t}{4+t^2}\,dt$

let $4+t^2 = k$

$2t\,dt = dk$

$t\,dt = \frac{dk}{2}$

$\int \frac{t}{4+t^2}\,dt$

$= \int \frac{1}{k}\,\frac{dk}{2}$

$= \frac{\log k}{2} + C$

$= \frac{\log(4+t^2)}{2} + C$

$\int_1^5 \frac{t}{4+t^2}\,dt = \frac{1}{2}\left[\log 29 - \log 5\right] = \frac{\log \frac{29}{5}}{2}$

Even funcⁿs: $f(x) = f(-x)$

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

e.g. x^2



odd funcⁿs: $f(x) = -f(x)$

$$\int_{-a}^{a} f(x)\,dx = 0$$

e.g. x

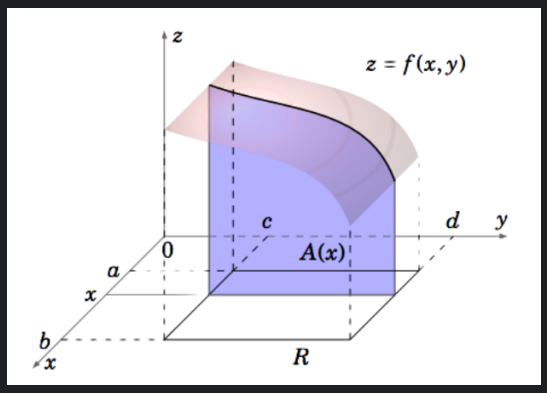


Double integrals (volume under the curve)

$z = f(x, y)$

$$V = \int_a^b \int_c^d f(x,y)\,dx\,dy$$

$$= \int_a^b \left[\int_c^d f(x,y)\,dy\right] dx$$



e.g. Find the volᵐ of the plane $z = 8x + 6y$ over the rectangle $[0, 1] \times [0, 2]$ $\quad 0 \le x \le 1$, $0 \le y \le 2$

→ $$V = \int_0^1 \left[\int_0^2 (8x + 6y)\,dy\right] dx$$

$$= \int_0^1 \left[8xy + \frac{6y^2}{2}\right]_{y=0}^{2} dx$$

$$= \int_0^1 \left(8x \cdot 2 + 6 \cdot \frac{4}{2}\right) dx$$

$$= \int_0^1 (16x + 12)\,dx$$

$$= \left[16\frac{x^2}{2} + 12x\right]_{x=0}^{1}$$

$$= \frac{16}{2} + 12$$

$$= 20$$

Q. $z = e^{x+y}$ $\quad 2 \le x \le 3, \quad 1 \le y \le 2$

$$\rightarrow \int_2^3 \int_1^2 e^{x+y}\, dy\, dx$$

$$\int e^{x+k}\, dx = e^{x+k}$$

$$= \int_2^3 \left[e^{x+y} \right]_{y=1}^{2} dx$$

$$= \int_2^3 (e^{x+1} - e^{x+2})\, dx$$

$$= \left[e^{x+1} \right]_{x=2}^{3} - \left[e^{x+2} \right]_{x=2}^{3}$$

$$= e^4 - e^3 - (e^5 - e^4)$$

$$= e^4 - e^3 - e^5 + e^4$$